Casey Barton

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PROFILE

I build the systems underneath agent deployments: the agent runtimes, tool and MCP integrations, eval harnesses, and observability that turn one-off field wins into reusable platform. Every agent I ship runs under an eval discipline first — golden sets, behavioral assertions, trace logging, judge criteria — and every production failure mode I have found has something built to catch it. Fail-closed by default.

Three years as a founder running an evidence-grade agent control plane from Honolulu: hand-rolled multi-agent loops with planning and tool-use, retrieval, memory, guardrails, and human handoffs; **65** MCP connectors exposing governed context to any MCP-compatible system; full run traces so any execution can be reconstructed; model-agnostic abstractions with benchmark loops that track cost, latency, and accuracy on every release.

CORE COMPETENCIES

Engineering: TypeScript · Python · Node.js · SQL · Bash · REST APIs · containers · GitHub Actions (CI/CD) · Supabase · Postgres · event-driven systems

AI & Evaluation: agent runtimes · MCP servers & connectors · LLM evaluation harnesses · judge-in-the-loop patterns · trace intelligence · failure-mode instrumentation · benchmarking & experiments · memory systems · RAG · function calling · structured outputs · hand-rolled multi-agent loops · guardrails · fail-closed recovery

Production: production platform engineering · observability & run logging · latency & token-cost benchmarks · CI regression gates · decision audit receipts · documentation engineers actually read

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Own the harness: eval suites, traces, and judge-in-the-loop patterns that tell whether an agent is actually doing the job, against a baseline. Prompt and configuration changes are validated by eval harnesses before anything touches production — an agent that cannot be measured cannot be trusted.
- Design agent systems for failure: instrument the failure modes found in production — stuck loops, bad retrievals, tool misuse, context blowouts — and build the catchers: replay loops, fail-closed gates, behavioral assertions, and recovery restricted to evidence declared before reassessment.
- Author **65** MCP connectors and integrations that expose tools and structured resources to any MCP-compatible system — the shared, governed context layer: ERPs, CRMs, data pipelines, webhooks — with evidence boundaries and audit receipts.
- Build model-agnostic: abstract the model layer across providers and benchmark when a new release lands — agent primitives tested against datasets, per-task latency and token-cost instrumentation, findings turned into configuration improvements.
- Run an evidence-grade agent control plane across a **1,395-repository** estate: planning and tool-use loops, retrieval over a domain ontology, memory, guardrails, typed handoff contracts, human

approval gates on consequential actions — backed by a **989-test** suite, **62/62** verification checks green.

- Turn field learning into reusable infrastructure: one-off fixes become shared patterns with written design docs, versioned eval suites, and the same governed context for every downstream tool.

EDUCATION

University of Hawaii at Manoa 2016
Bachelor of Science

AWS Cloud Institute 2025
Cloud Computing program

Colorado State University
Undergraduate coursework

SELECTED PROOF

Job-App Helix — evaluation-gated campaign engine

- Attributable campaign engine with immutable evidence models, stable finding codes, and fail-closed **GO / NO-GO** decisions; recovery restricted to contingency evidence declared before reassessment.
- Verification gates on every agent action: no agent completion claim without independent provider readback — the eval gate decides, not the agent.

GlacierEQ estate — production agent systems

- Durable patterns for claim tracking, evidence boundaries, audit receipts, and human-in-the-loop control across the full estate.
- Per-task latency instrumentation and decision audit logs across agent runs — the tracing practice that makes benchmarking honest.

SKILLS

Languages: TypeScript · Python · SQL · Bash · JavaScript / Node.js

AI / Evaluation: agent runtimes · MCP servers & connectors · eval harnesses · judge-in-the-loop · trace intelligence · failure-mode instrumentation · benchmarking · memory systems · guardrails · typed handoff contracts

Production: platform engineering · observability & run logging · CI regression gates · fail-closed recovery · cost/latency tradeoffs · documentation & handover

Evidence boundary: portfolio repositories are independent work. Domain-inspired demonstrations are not claims of company affiliation, production deployment, or certification.